

1.5 Moments of Continuous R.V.

(a) **Raw moments** : If X is a continuous r.v. with p.d.f. $f(x)$ then raw moment is denoted by μ'_r and it is defined as,

$$\mu'_r = E(X^r) = \int_{-\infty}^{\infty} x^r f(x) dx$$

provided, $\int_{-\infty}^{\infty} |x|^r f(x) dx < \infty$.

Therefore, the first four raw moments will be,

$$\mu'_1 = \int_{-\infty}^{\infty} x f(x) dx = \text{mean}, \mu'_2 = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$\mu'_3 = \int_{-\infty}^{\infty} x^3 f(x) dx \text{ and } \mu'_4 = \int_{-\infty}^{\infty} x^4 f(x) dx.$$

(b) **Central moments** : If $E(X) = m$ then r^{th} central moment is given by,

$$\mu_r = E(X - m)^r = \int_{-\infty}^{\infty} (x - m)^r f(x) dx$$

provided, $\int_{-\infty}^{\infty} |(x - m)^r| f(x) dx < \infty$

Clearly,

$$\mu_1 = E(X - m) = 0, \mu_2 = E(X - m)^2 = \text{Var}(X)$$

$$\mu_3 = E(X - m)^3 \text{ and } \mu_4 = E(X - m)^4$$

Relation between central moments and raw moments

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_3 = \mu_3' - 3 \mu_2' \mu_1' + 2 (\mu_1')^3$$

$$\mu_4 = \mu_4' - 4 \mu_2' \mu_1' + 6 \mu_2' (\mu_1')^2 - 3 (\mu_1')^4$$

(c) Moments about 'a' will be

$$\mu_r(a) = E(x-a)^r = \int_{-\infty}^{\infty} (x-a)^r f(x) dx, r = 1, 2, 3, \dots$$

If particular for $a = 0$ we get, raw moments and $a = E(X)$ we get central moment.

Pearsonian coefficients :

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}, \gamma_1 = \sqrt{\beta_1} = \mu_3 / \sqrt{\mu_2^3}$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2}, \gamma_2 = \beta_2 - 3.$$

Harmonic mean (H) is given by,

$$\frac{1}{H} = E\left(\frac{1}{X}\right) = \int_{-\infty}^{\infty} \frac{1}{x} f(x) dx \quad \frac{1}{E(X)}$$

If X takes positive values only then, H is defined.

Geometric mean (G) is given by,

$$G = \text{Antilog } E(\log X) = \text{Antilog } \int_0^{\infty} \log x f(x) dx$$

(where, X is a r.v. taking non-negative values.)

Mean Deviation (M.D.) about mean (m) is given by

$$\text{M.D.} = E[|X - m|] = \int_{-\infty}^{\infty} |x - m| f(x) dx$$

1.6 Moment Generating Function (M.G.F.)

(Oct. 2)

M.G.F. is useful in many ways in the study of probability distribution, such as,

- (i) to find moments.
- (ii) to find probability distribution of $g(x)$; a function of r.v. X .
- (iii) to verify whether the random variables are independent in multivariate probability distribution.

Definition : If X is a continuous r.v. with p.d.f. $f(x)$ then moment generating function of X is denoted by $M_X(t)$ and defined as, (Oct. 2)

$$M_X(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

provided the integral exists for some h such that $-h < t < h, h > 0$

Note : $M_X(t)$ can be expressed as power series in t as follows :

$$M_X(t) = E(e^{tx}) = E\left(1 + tx + \frac{t^2 x^2}{2!} + \dots\right)$$

$$M_X(t) = 1 + \mu'_1 t + \mu'_2 \cdot \frac{t^2}{2!} + \mu'_3 \cdot \frac{t^3}{3!} + \dots$$

Raw moments using M.G.F.

Method 1 : $\mu'_r = r^{\text{th}}$ raw moment

= coefficient of $\frac{t^r}{r!}$ in the expansion of $M_X(t)$.

Method 2 : $\mu'_r = \left[\frac{d^r M_X(t)}{dt^r} \right]_{t=0}$

7 Properties of M.G.F.

(1) $M_X(0) = 1$

(2) Effect of change of origin and scale.

Result : If X is a r.v. with M.G.F. $M_X(t)$

then, (i) $M_{X+a}(t) = M_X(t) \cdot e^{at}$

(ii) $M_{cX}(t) = M_X(ct)$

(iii) $M_{a+cX}(t) = e^{at} M_X(ct)$, where a, c are constants.

Proof : (i) $M_{X+a}(t) = E[e^{(X+a)t}] = E[e^{tX} \cdot e^{at}] = e^{at} M_X(t)$

(ii) $M_{cX}(t) = E[e^{cX \cdot t}] = E[e^{X(ct)}] = M_X(ct)$

(iii) $M_{a+cX}(t) = E[e^{(a+cX)t}] = E[e^{at + cXt}]$
 $= e^{at} E[e^{cXt}] = e^{at} M_X(ct)$

(3) If X and Y are independent r.v.s. then,

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$$

Proof : $M_{X+Y}(t) = E[e^{t(X+Y)}]$
 $= E(e^{tX} \cdot e^{tY}) \quad (\because X \text{ and } Y \text{ are independent})$
 $= E(e^{tX}) \cdot E(e^{tY})$
 $= M_X(t) \cdot M_Y(t).$

Note : $M_{aX+bY+c}(t) = e^{ct} M_X(at) M_Y(bt).$

(4) **Uniqueness property :** For a given p.d.f. there is unique M.G.F.
and for a given M.G.F. there is unique p.d.f.

1.8 Cumulant Generating Function (C.G.F.)

In order to obtain M.G.F. we have used the transformation e^{tx} , what do we get, if we use inverse transformation $\log_e$ may be a question of interest. Accordingly, $\log_e M_X(t)$ is obtained. It is called as cumulant generating function, which is found to be useful to find the central moments easily.

Definition : If X is a r.v. with M.G.F. $M_X(t)$ then $\log_e M_X(t)$ is called as cumulant generating function. It is denoted by $K_X(t)$.

Thus, $K_X(t) = \log_e M_X(t)$. Like M.G.F., $K_X(t)$ can also be expressed as a power series in t .

$$K_X(t) = k_1 t + k_2 (t^2/2!) + k_3 (t^3/3!) + \dots$$

$\therefore k_r = r^{\text{th}}$ cumulant = Coefficient of $\frac{t^r}{r!}$ in the expansion of $K_X(t)$

We can also get k_r by successive differentiation of $K_X(t)$.

$$k_r = \left[\frac{d^r K_X(t)}{dt^r} \right]_{t=0}$$

We summarise below the relation between cumulants and moments

$$k_1 = \mu'_1, k_2 = \mu_2, k_3 = \mu_3, k_4 = \mu_4 - 3 \mu_2^2$$

Thus, we get first four central moments of X as follows.

$$\mu_1 = 0, \mu_2 = k_2, \mu_3 = k_3, \mu_4 = k_4 + 3 k_2^2.$$

Properties of C.G.F. :

(1) If a and b are constants then,

$$K_{X+a}(t) = at + K_X(t)$$

$$K_{a+bX}(t) = at + K_X(bt)$$

(2) If X and Y are independent random variables then,

$$K_{X+Y}(t) = K_X(t) + K_Y(t)$$

1.9 Median and Quartiles

If X is a continuous r.v. with p.d.f. $f(x)$ then *median* is a point which divides the area under the curve $y = f(x)$ into two equal parts. Therefore, *median* (μ) is a value of r.v. X such that

$$P(X \leq \mu) = P(X \geq \mu) = \frac{1}{2}$$

$$\therefore \int_{-\infty}^{\mu} f(x) dx = \int_{\mu}^{\infty} f(x) dx = \frac{1}{2}$$

$$\text{i.e.} \quad F(\mu) = 1 - F(\mu) = \frac{1}{2}$$

$$\therefore F(\mu) = \frac{1}{2}$$

where $F(\cdot)$ is a distribution function.

Quartiles of r.v. X or probability distribution can be defined on similar lines.

First Quartile : The first quartile Q_1 is that value of r.v. X such that

$$\therefore P(X \leq Q_1) = \frac{1}{4} \text{ or } P(X \geq Q_1) = \frac{3}{4}$$

$$\text{i.e.} \quad F(Q_1) = \frac{1}{4}$$

Second Quartile : It is same as median.

Third Quartile : The third quartile Q_3 is that value of r.v. X such that

$$P(X \leq Q_3) = \frac{3}{4} \text{ or } P(X \geq Q_3) = \frac{1}{4}$$

$$\text{i.e.} \quad F(Q_3) = \frac{3}{4}$$

Note : The concept of quartiles can be extended to other partition values such as deciles and percentiles.

Deciles : The i^{th} decile (D_i) is a value of r.v. X such that,

$$P(X \leq D_i) = i/10; \quad i = 1, 2, \dots, 9.$$

$$\text{i.e.} \quad F(D_i) = i/10$$

Percentiles : The i^{th} percentile (P_i) is a value of r.v. X such that,

$$P(X \leq P_i) = i/100; \quad i = 1, 2, \dots, 99.$$

$$\text{i.e.} \quad F(P_i) = i/100.$$

1.10 Mode

Definition : Mode is a value of X at which $f(x)$ is maximum.

In case of a continuous random variable if p.d.f. $f(x)$ is a differentiable function, then mode is that value of x which satisfies the following conditions, $\frac{df(x)}{dx} = f'(x) = 0$ and $\left[\frac{d^2f(x)}{dx^2} = f''(x) \right]_{x=a} < 0$